THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 0	Common difference is -1; $a_{p+q}=0$
2	**(d)** 8	$(m+n)^2-m^2$
3	**(a)** 12	Let the point be $(0,y)$
4	**(b)** 54	$a_n=\frac{2S_n}{n}-a$
5	**(d)** 6	Find b by Pythagoras
6	**(b)** 63	Twice the area
7	**(d)** 46	$a_n=S_n-S_{n-1}$
8	**(b)** $\frac{12}{5}$	$\frac{1}{2}ab=\frac{1}{2}ch$
9	**(d)** $\frac{16}{3}$	Equate squared distances
10	**(d)** 4	$d=2p$
11	**(d)** 14	Sides in ratio $a:b$
12	**(b)** 6	$x=0$: $mc+na=0$
13	**(a)** Both A and R are true and R is the correct explanation of A.	$7-3=11-7=15-11=4$; R is the defining property of an AP.
14	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{AD}{DB}=\frac{AE}{EC}\Rightarrow\frac{2}{3}=\frac{4}{EC}\Rightarrow EC=6$; R is the Basic Proportionality Theorem.
15	**(a)** Both A and R are true and R is the correct explanation of A.	$\sqrt{9+16}=5$; R is the formula used.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$a+3d=9$, $a+8d=4$ (1); $d=-1$, $a=12$ so $a_{13}=0$ (1).	1+1
17	$\triangle ADE\sim\triangle ABC$ (1); $\frac{DE}{BC}=\frac{AD}{AB}=\frac{3}{7}$ (1); $DE=6$ cm (1).	1+1+1
18	$AB^2=25$, $BC^2=25$, $CD^2=25$, $DA^2=25$ (2); all four sides equal (1); diagonals $AC^2=80$, $BD^2=20$ differ so it is a rhombus, not a square (2).	2+1+2
19	$a_1=S_1=7$ (1); $S_2=20$, $a_2=13$, $d=6$ (1).	1+1
20	Ratio of areas = (ratio of sides)^2 (1); $\frac{27}{75}=\frac{3^2}{x^2}$ (1); $x=5$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\frac{6}{2}(2a+5d)=75$ and $\frac{10}{2}(2a+9d)=185$ (1); simplify to linear equations in a,d (1); $a=5$, $d=3$, AP $5,8,11,14,\ldots$ (1).	1+1+1
22	$\triangle AOB\sim\triangle COD$ (AA) (1); $\frac{AO}{OC}=\frac{AB}{DC}=2:3$ (1); area ratio $=4:9$ (1).	1+1+1
23	$AB^2=25$, $BC^2=25$, $CD^2=25$, $DA^2=25$ (2); all four sides equal (1); diagonals $AC^2=90$, $BD^2=10$ differ so it is a rhombus, not a square (2).	2+1+2
24	$d=4$, $a=6$ (2); $S_{16}=576$ (1).	2+1
	OR: Altitude bisects the base: $h^2=(6)^2-3^2=27$ (1+1); $h=3\sqrt{3}$ cm (1).	1+1+1
25	P divides AB in $1:2$: $(3,\ 2)$ (1.5); Q divides AB in $2:1$: $(6,\ 4)$ (1.5).	1.5+1.5
	OR: Terms form an AP with $d=7$ (1); count and endpoints (1); sum $=2916$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $a=100$, $d=20$ (1). (ii) $a_{10}=100+9\times20=₹280$ (1). (iii) $S_{12}=\frac{12}{2}[2(100)+11(20)]=6\times420=₹2520$ (1). (iv) $100+(n-1)20=400\Rightarrow n=16$ (1); $S_{16}=\frac{16}{2}(100+400)=₹4000$ (1).	1+1+1+2

27	Centroid $= (6,\ \frac{8}{3})$ (2); area $= \frac{1}{2}\lvert \dots \rvert = 48$ sq. units (3).	2+3
	OR: AP: $a = 2500$, $d = 150$, $n = 18$ (1); last $= a + (n-1)d = 5050$ (2); total $= S_n = \frac{18}{2}(2500 + 5050) = 67950$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 11	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
2	**(d)** 25	Side $= s$
3	**(b)** 28	Use the area formula
4	**(d)** 0	Common difference is -1; $a_{p+q} = 0$
5	**(b)** 21	$(m+n)^2 - m^2$
6	**(a)** 9	Let the point be $(0, y)$
7	**(d)** 59	$a_n = \frac{2S_n}{n} - a$
8	**(b)** 30	Find b by Pythagoras
9	**(c)** 44	Twice the area
10	**(a)** 54	$a_n = S_n - S_{n-1}$
11	**(d)** $\frac{60}{13}$	$\frac{1}{2}ab = \frac{1}{2}ch$
12	**(b)** 4	Equate squared distances
13	**(c)** A is true but R is false.	$a_{10} = 2 + 9 \times 3 = 29$, so A is true; the correct formula is $a + (n-1)d$, so R is false.
14	**(c)** A is true but R is false.	$AC^2 = 36 + 64 = 100$, so $AC = 10$: A true; the theorem holds for every right-angled triangle, so R is false.
15	**(c)** A is true but R is false.	$\left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = (5, 1)$: A true; the formula uses sums, not differences, so R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(8-k) + 2(k-5) + 4(5-8)\rvert = 0$ (1); $k = 14$ (1).	1+1+1
17	$2a + 8d = 34$, $2a + 9d = 37$ (1); $d = 3$, $a = 5$ (1).	1+1
18	Hypotenuse 5 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{12}{5}$ cm (1).	1+1
19	Midpoint of the opposite side $= (3,\ 4)$ (1); length $= 5$ (1).	1+1
20	$a + 4d = 12$, $a + 11d = 5$ (1); $d = -1$, $a = 16$ so $a_{17} = 0$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$AC = 5$ cm (1); $AD = \frac{AB^2}{AC} = \frac{9}{5}$ cm, $DC = \frac{BC^2}{AC} = \frac{16}{5}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{12}{5}$ cm (1).	1+1+1
22	$(6)^2 + (y-3)^2 = 100$ (1); $(y-3)^2 = 64$ (1); $y = 3 \pm 8$, i.e. $y = 11$ or -5 (1).	1+1+1
23	$\frac{5}{2}(2a + 4d) = 50$ and $\frac{9}{2}(2a + 8d) = 162$ (1); simplify to linear equations in a, d (1); $a = 2$, $d = 4$, AP $2, 6, 10, 14, \ldots$ (1).	1+1+1
24	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 3:5$ (1); area ratio $= 9:25$ (1).	1+1+1
	OR: $AB^2 = 100$, $BC^2 = 100$, $CD^2 = 100$, $DA^2 = 100$ (2); all four sides equal (1); diagonals $AC^2 = 320$, $BD^2 = 80$ differ so it is a rhombus, not a square (2).	2+1+2
25	$d = 5$, $a = 9$ (2); $S_{20} = 1130$ (1).	2+1
	OR: Altitude bisects the base: $h^2 = (8)^2 - 4^2 = 48$ (1+1); $h = 4\sqrt{3}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) AA (each has a right angle and the same sun-angle) (1). (ii) $\frac{h}{24} = \frac{1.5}{2} \Rightarrow h = 18$ m (1). (iii) $\frac{6}{s} = \frac{1.5}{2} \Rightarrow s = 8$ m (1). (iv) $\sqrt{18^2 + 24^2} = \sqrt{900} = 30$ m by the Pythagoras theorem (2).	1+1+1+2
27	AP with $a = 90$, $l = 30$, $n = 31$ (1); $S_n = \frac{31}{2}(90 + 30) = 1860$ cm (2); middle term $= \frac{90+30}{2} = 60$ cm (2).	1+2+2

	OR: Right triangle with legs 50 and 120 (1); distance $= \sqrt{2500 + 14400} = 130$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 1116	Find n first
2	**(d)** $\frac{25}{13}$	$AB^2 = AD \cdot AC$
3	**(c)** 50	$s^2 = d^2/2$
4	**(a)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
5	**(b)** 49	Side $= s$
6	**(c)** 28	Use the area formula
7	**(b)** 0	Common difference is -1; $a_{p+q} = 0$
8	**(c)** 40	$(m+n)^2 - m^2$
9	**(c)** $\frac{22}{3}$	Let the point be $(0, y)$
10	**(d)** 52	$a_n = \frac{2S_n}{n} - a$
11	**(a)** 60	Find b by Pythagoras
12	**(c)** 78	Twice the area
13	**(d)** A is false but R is true.	Differences are $3, 5, 7, \ldots$, which are not constant: A false; R is true.
14	**(d)** A is false but R is true.	$5^2 + 6^2 = 61 \neq 49 = 7^2$, so it is not right-angled: A false; R is the (true) converse of the Pythagoras theorem.
15	**(d)** A is false but R is true.	$(0, 5)$ has x-coordinate 0 so it is on the y-axis: A false; R is true.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{2}{7}$ (1); $DE = 6$ cm (1).	1+1+1
17	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
18	$a_1 = S_1 = 7$ (1); $S_2 = 18$, $a_2 = 11$, $d = 4$ (1).	1+1
19	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{32}{162} = \frac{4^2}{x^2}$ (1); $x = 9$ cm (1).	1+1+1
20	Area of triangle must be 0 (1); $\frac{1}{2}\lvert -2(7-k) + 0(k-3) + 3(3-7)\rvert = 0$ (1); $k = 13$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	P divides AB in $1:2$: $(6, 5)$ (1.5); Q divides AB in $2:1$: $(9, 7)$ (1.5).	1.5+1.5
22	Terms form an AP with $d = 9$ (1); count and endpoints (1); sum $= 4867$ (1).	1+1+1
23	$AC = 13$ cm (1); $AD = \frac{AB^2}{AC} = \frac{25}{13}$ cm, $DC = \frac{BC^2}{AC} = \frac{144}{13}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{60}{13}$ cm (1).	1+1+1
24	$(12)^2 + (y-1)^2 = 169$ (1); $(y-1)^2 = 25$ (1); $y = 1 \pm 5$, i.e. $y = 6$ or -4 (1).	1+1+1
	OR: $\frac{7}{2}(2a + 6d) = 98$ and $\frac{12}{2}(2a + 11d) = 228$ (1); simplify to linear equations in a, d (1); $a = 8$, $d = 2$, AP $8, 10, 12, 14, \ldots$ (1).	1+1+1
25	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 1:4$ (1); area ratio $= 1:16$ (1).	1+1+1
	OR: $AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$ (1). (ii) $SL = \sqrt{6^2 + 8^2} = 10$ units (1). (iii) $LH = \sqrt{(8-2)^2 + 0} = 6$ units (1). (iv) $SH = 8$; $SH^2 + LH^2 = 64 + 36 = 100 = SL^2$, so by the converse of Pythagoras the triangle is right-angled at H (2).	1+1+1+2

27	Draw a horizontal line from the top of the shorter pole: right triangle with legs 27 and 36 (2); $x^2 = 729 + 1296$ (1); $x = 45$ m (2).	2+1+2
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{11}{2}, 7\right)$ (2); this is also the midpoint of BD (1); $D = (4,\ 11)$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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